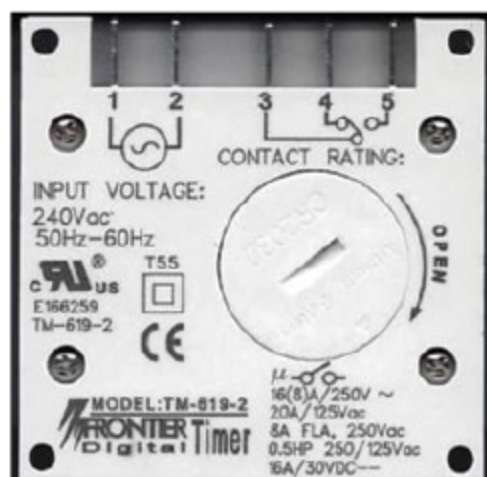


The datasheet and manual are included in this month's EFY DVD and are also available for free download at source.efymag.com



(a) Front view



(b) Rear view

Fig. 2: Digital timer programmable time switch TM-619

both IC2 and IC3 are high. Therefore output pin 3 of IC4 (AND gate N1) is high, which is connected to trigger input A2 of IC5. Here IC5 is used as a falling-edge-triggered one-shot monostable multivibrator for pulse width of 3.2 seconds by choosing the values of resistor R5 and capacitor C9 as 10-kilo-ohm and 470 μ F, respectively. These values are selected depending on the time duration for which the bell should ring. You can use a potentiometer in place of the fixed-value R5 to vary the time duration as per the time slot of each class.

Normally, when $\bar{Q}$ output of IC2 and IC3 are high, N1 gate output of IC4 is also high. In this situation, Q output (pin 6) of IC5 is low. As transistor T1 is non-conducting, relay RL1 does not energise and the bell connected across CON2 remains off. The bell rings when Q output of IC5 goes high for predetermined time duration of 3.2 seconds. LED1 glows during this time duration.

For programming digital timer switch TM-619, refer its user manual on the Internet or refer this month's EFY

TYPICAL SCHOOL TIME-TABLE AND TIMER STATUS

Sr. No	Time	Sequence of period/lecture	Status of timer output (TM-619)
1)	7:30am	Period 1	ON
2)	8:30am	Period 2	OFF
3)	9:30am	Period 3	ON
4)	10:00am	Break (starts)	OFF
5)	10:45am	Break (ends)	ON
6)	11:45am	Period 4	OFF
7)	12:45pm	Period 5	ON
8)	13:45pm	End of Lectures	OFF

DVD. Working of the circuit with timer switch TM-619 is given below:

1. Program the digital timer switch (TM-619) as per the required timetable. As an example, typical timetable of a college/school is shown in the table. Note that at the end time of the lecture, the timer should be in 'off' condition as shown in the table. If the sequence ends with 'on' time, put an additional 'off' time by adding 5-minute time interval. This ensures that timer TM-619 output is in off condition so that it begins the next cycle with 'on' condition. Note that whenever the timer output toggles between 'on' and 'off' conditions, output pin 6 of IC5 goes high for 3.2 seconds. This means, the AC bell rings for 3.2 seconds.

2. After programming, put the timer in auto mode.

3. When the timer clock reaches preset

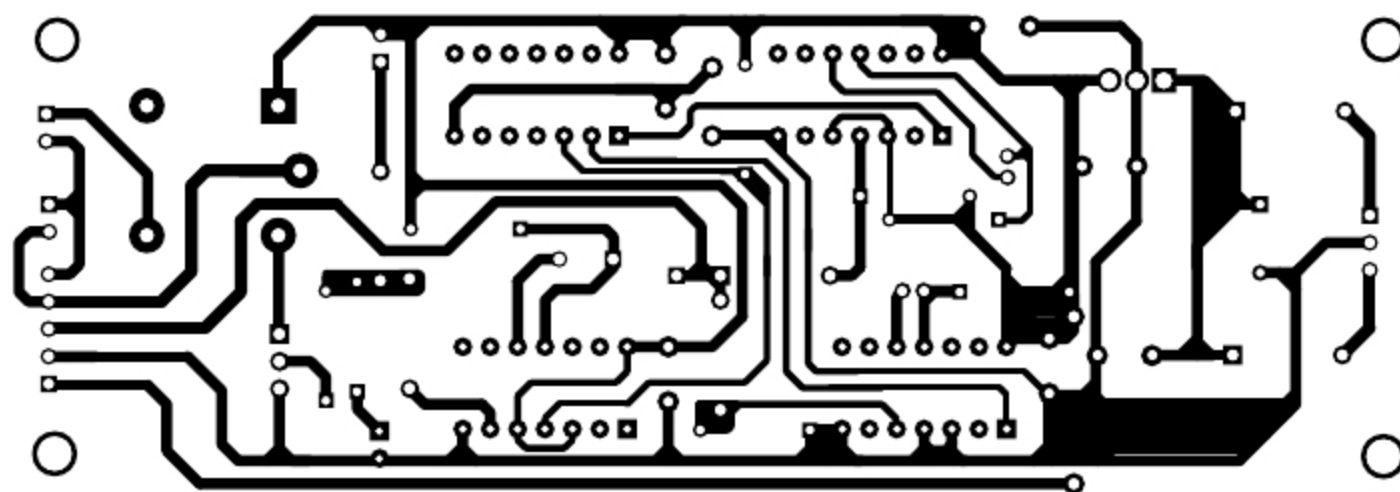


Fig. 3: Actual-size PCB layout of the programmable automatic bell system

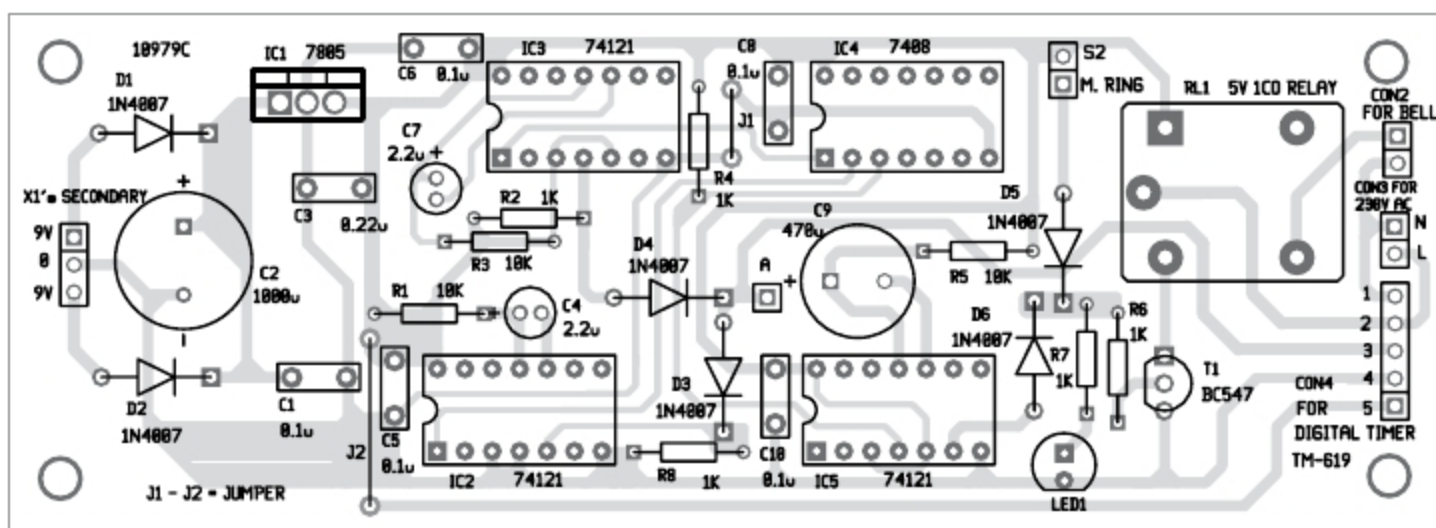


Fig. 4: Components layout for the PCB